Client Layer (frontend/src)
User Interfaces & Pages (pages/)
Projects & Workspace pages/projects
Login & Guest Auth pages/login
Upload & Previews pages/upload
Validation Workspace pages/validation
Mobile Capture via QR pages/upload/MobileCapture
Components & State (components/, hooks/, utils/)
PDF Page Splitter pdfjs-dist, HTML5 Canvas
Validation Workspace ValidationWorkspace
Camera photo upload
Frontend Services (services/)
uploadService.ts
llmService.ts
projectService.ts
documentService.ts
export/ (batchExportService, templates, writers)
supabaseClient.ts (Anon Key)
Auth Guard & Session context/AuthProvider
Table Editor & Undo/Redo useTableEditor, useUndoRedo
OCR Issue Scanner useReviewQueue, useReviewScanner
OCR Progress Tracker context/OcrProgressProvider
Session Cookie
Bearer JWT
Bearer JWT + Multipart
API Gateway & Routing (backend/src)
Express Routers (routes/)
/api/upload upload.ts (Legacy/Guest)
/api/extraction extractionRoutes.ts
/api/llm llmRoutes.ts
/api/projects projectRoutes.ts
/api/documents documentRoutes.ts
/api/qr qr.ts
Session & Multer express-session, multer
Auth Middleware middleware/auth.ts (JWT verify)
Controllers (controller/)
extraction.ts
documents.ts
projects.ts
qrController.ts
StructuredDocument (Blocks, Tables, Fields)
StructuredDocument
upload.ts
llm.ts
Server Entrypoint backend/app.ts + src/app.ts
Auth & Token
Domain Logic & Processing (backend/src/services)
OCR Coordinator ocr/ocr.ts + utils/analyseBuffer.ts Azure DI Client, Retries & Buffer Prep
Database Client supabaseClient.ts (Service Role)
QR Session Store qrService.ts
Layout Analysis Output
Document Structuring Pipeline ocr/pipeline/pipeline.ts Profile Registry: table, receipt, invoice, generic
Sequential Claiming
Preprocess Image Buffer
Pipeline Extractors ocr/pipeline/extractors.ts nativeTableExtractor, plainTextExtractor, llmStructurer
Raw Layout & Geometry
Bypass RLS Queries
Indent & Virtual Row Hierarchy
Classification & Floating Table Extraction
Image Processor imageProcessor.ts (Sharp + OpenCV) Resize & Quality Checks (Blur/Glare/Contrast)
Hierarchy & Layout Engine ocr/pipeline/hierarchy.ts + layout.ts Indentation, Parent-Child Rows & Polygon Union
OCR Preprocessor ocrPreprocessor.ts (Sharp + OpenCV) Unsharp Masking (Gaussian Blur + Weighted Diff) & Grayscale
LLM Service llm/aiService.ts (Google GenAI) Chat, Cell & Bulk Corrections, Format Profiling
Optimized PNG Buffer
Sharpened Grayscale Buffer
Field Corrections, Format Profiling & Chat
Object Storage Client r2Client.ts (AWS S3 SDK)
Local Disk Cache
Databases, Storage & Third-Party APIs
Cognitive & AI Cloud Services
Azure Document Intelligence prebuilt-layout model
Google Gemini LLM Gemini Flash Lite (gemini-flash-lite-latest & gemini-3.5-flash-lite)
Supabase Platform
PostgreSQL Database projects, documents, document_pages Row Level Security (RLS)
Put / Get Presigned URL
Enforce Tenant RLS
Storage Infrastructure
Cloudflare R2 Bucket PNG Page Blobs: {user}/{project}/{doc}/page-{n}.png
Local Disk /uploads Ephemeral session uploads & QR temp files
Supabase Auth Users, JWT & OAuth